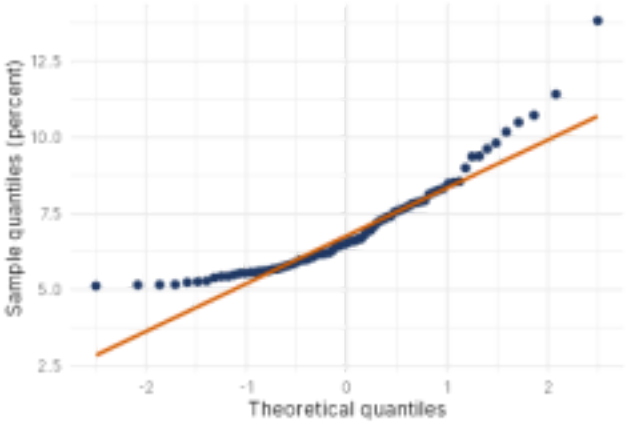
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

R.No: 2 Reg. No: 192424331 Name: AKILA S

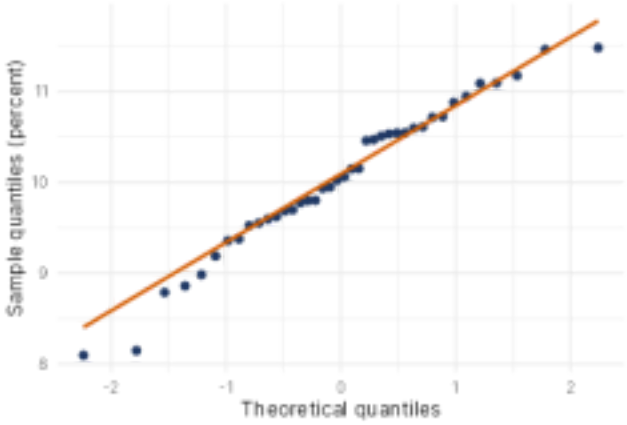
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: battery drain per hour (percent), n = 80



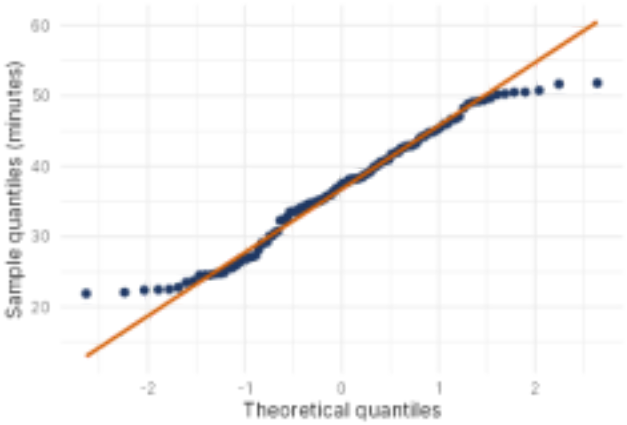
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

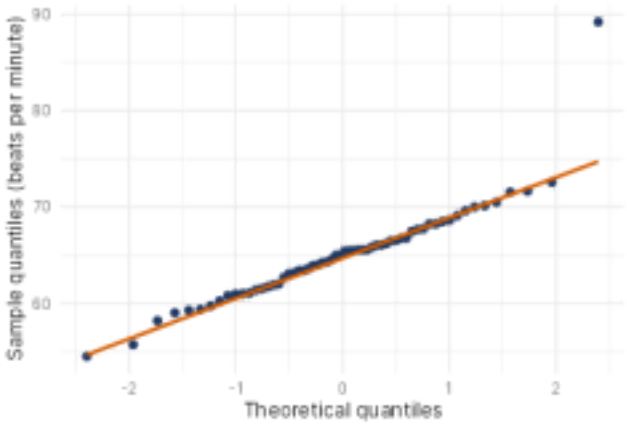
Chart 2: battery drain per hour (percent), n = 40



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: commute time (minutes), n = 120Chart 4: resting pulse (beats per minute), n = 60





Part B (4 marks)

Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

B1. Chart 4 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

SOLUTION:

Part A – Identify the Shape of Each Q-Q Plot

Chart 1: Battery Drain per Hour (n = 80)

Normal: No

Shape: Right-skewed

Evidence in the plot: The points bend away from the red reference line, particularly in the upper tail. The highest values lie well above the line, reaching approximately 13.8%, indicating a longer right tail.

Chart 2: Battery Drain per Hour (n = 40)

Normal: Approximately yes

Shape: Normal

Evidence in the plot: Most points lie close to the red reference line and follow an approximately straight pattern. Some deviations occur at the lower tail, but the overall pattern is reasonably consistent with a normal distribution.

Chart 3: Commute Time (n = 120)

Normal: No

Shape: Light-tailed

Evidence in the plot: The points at both extremes bend inward toward the centre relative to the red reference line. The upper-tail points fall below the line, while the lower-tail points lie above it, suggesting lighter tails than a normal distribution.

Chart 4: Resting Pulse (n = 60)

Normal: No

Shape: Normal with outlier(s)

Evidence in the plot: Most points follow the red reference line, but one upper-tail point lies far above it at approximately 89 beats per minute. This suggests an unusually high observation or upper-tail outlier.

Part B – Short Answers

B1. Chart 4 departs from normality. Name the departure and state the remedy.

Departure: Normal distribution with an upper-tail outlier.

Remedy: Investigate the unusually high value to check for measurement or recording errors. If the value is valid, use robust methods such as the median and interquartile range instead of relying only on the mean and standard deviation.

B2. Complete the R code to draw a Q-Q plot with its reference line.

```
ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```

Final Answer Summary

Chart	Normal?	Shape
Chart 1	No	Right-skewed
Chart 2	Approximately yes	Normal
Chart 3	No	Light-tailed
Chart 4	No	Normal with outlier(s)